CS-5136
Graph Theory

Day 11: Trees

From "what is a tree?" all the way to the Wiener index — built from the ground up

Where this fits

Your Mid-Semester exam stopped at Hamiltonian graphs (page 194 of the course slides). This deck starts exactly where that left off: the short "Centrality" section (distance, eccentricity, center, radius — pages 195–200) that you haven't been tested on yet, and then the full **Tree** chapter (pages 201–230), which is everything currently in your slides on trees. This matches Lesson-Plan Day 10, *"Tree, terminologies and properties"*, from your official syllabus. Spanning trees, counting trees, fundamental circuits and cut-sets (Lesson-Plan Days 12–15) are a later, separate chapter — not covered here.

Every definition is self-contained — nothing from outside this document is assumed. Every **if-and-only-if** statement is proved in *both* directions, explicitly labelled. Proofs are built up from the course slides' own sketches, completed in full using Narsingh Deo's *Graph Theory with Applications to Engineering and Computer Science* (Chapter 3) and Robin J. Wilson's *Introduction to Graph Theory* (Chapter 3) wherever the slides leave a step out.

Sources: A. Bhattacharjee, CS-5136 slides (pp. 195–230) • N. Deo, *Graph Theory...* Ch. 3, §§3-1–3-5 • R. J. Wilson, *Introduction to Graph Theory* (5th ed.), Ch. 3, Theorem 3.1

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How to Use This Deck

- Read every **Definition** box before the theorem that uses it — trees have a lot of new vocabulary, and the proofs lean on it hard.
- Every **Theorem** box is followed by a full **Proof**. Where the statement is an "if and only if", the proof is explicitly split into **Direction 1** and **Direction 2** (or, for the six-way equivalence in Section 3, into six separately labelled implications) — so you can see exactly which half you are reading.
- **Worked Examples** use small, concrete trees you can redraw on paper in ten seconds — use them to build intuition before trusting the general proof.
- **Notes** flag common exam traps: statements that sound true but aren't, or conditions students forget to check.

1 Before We Start: Distance, Eccentricity, and Centers

This short section covers the last bit of "general graph" material before we specialise to trees: the idea of *distance* between two vertices, and what it means for a vertex to be "in the middle" of a graph. We need this *before* we can talk about the center of a tree in Section 6.

Definition (Metric)

A function $f(x, y)$ of two variables is called a **metric** if it satisfies all three of:

1. **Non-negativity:** $f(x, y) \geq 0$, and $f(x, y) = 0$ if and only if $x = y$.
2. **Symmetry:** $f(x, y) = f(y, x)$.
3. **Triangle inequality:** $f(x, y) \leq f(x, z) + f(z, y)$ for every z.

This is exactly the same list of rules that ordinary straight-line distance satisfies in the plane — it's what lets us call something "a distance" at all.

Definition (Distance in a Graph)

For a connected graph G and two vertices $u, v \in V(G)$, the **distance** $d(u, v)$ is the length (number of edges) of a *shortest* u–v path in G.

Theorem — Distance Is a Metric

In a connected graph G, the function $d(u, v)$ is a metric.

Proof. We check all three conditions.

Non-negativity. A path has ≥ 0 edges, so $d(u, v) \geq 0$ always. If $u = v$, the shortest "path" is the trivial path of length 0, so $d(u, v) = 0$. Conversely if $d(u, v) = 0$ then the shortest u–v walk uses 0 edges, which is only possible if $u = v$.

Symmetry. Reversing a u–v path of length ℓ gives a v–u path of the same length ℓ, and this reversal is a bijection between u–v paths and v–u paths. So the *shortest* length is the same read either way: $d(u, v) = d(v, u)$.

Triangle inequality. Let z be any third vertex. Take a shortest u–z path (length $d(u, z)$) and a shortest z–v path (length $d(z, v)$), and concatenate them at z. This gives a u–v *walk* of length $d(u, z) + d(z, v)$. A shortest u–v *path* can be extracted from any u–v walk by deleting repeated vertices (deleting a repeated vertex only removes edges, never adds them), so

$$d(u, v) \;\leq\; \text{length of this walk} \;=\; d(u, z) + d(z, v).$$

■

Definition (Eccentricity, Center, and Radius)

Let G be a connected graph and $v \in V(G)$.

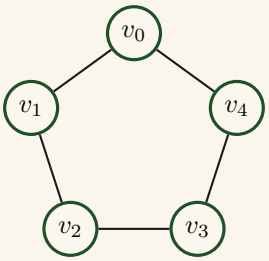
- The **eccentricity** of v, written $E(v)$, is the distance from v to the *farthest* vertex in G:

$$E(v) = \max_{x \in V(G)} d(v, x).$$

- A vertex with the *minimum* eccentricity in G is called a **center** of G.
- The eccentricity of a center is called the **radius** of G, written $\text{rad}(G)$.
- The **diameter** of G, written $\text{diam}(G)$, is the *maximum* eccentricity over all vertices — equivalently, the length of the longest shortest-path in G.

Example

Take the 5-cycle C_5 below. By symmetry every vertex sees one neighbour at distance 1 and two others at distance 2, so $E(v) = 2$ for *every* vertex. Every vertex is therefore a center, and $\text{rad}(C_5) = \text{diam}(C_5) = 2$.



Note

A graph can have *many* centers — a cycle is the extreme case where every single vertex is one. Trees behave very differently: Section 6 shows a tree has *at most two* centers, and if there are two, they must be adjacent.

Other Notions of Centrality — Just the Names

Eccentricity-based centrality isn't the only way to ask "which vertex matters most?" Two more you should recognise by name (not examined in depth here):

- **Degree centrality** — just the degree of the vertex; the simplest possible measure.
- **Geodesic (betweenness) centrality** — the fraction of all shortest paths (between *other* pairs of vertices) that pass through v.

Other named variants (closeness centrality, eigenvector centrality, PageRank centrality) exist but are outside this course.

Theorem — A Diameter Trade-off Between G and $\overline{G}$

If G is a simple graph with $\text{diam}(G) \geq 3$, then $\text{diam}(\overline{G}) \leq 3$, where $\overline{G}$ denotes the complement of G.

Proof. Suppose $\text{diam}(G) \geq 3$. Then there exist two vertices u, v with $d_G(u, v) \geq 3$. In particular u, v are non-adjacent in G (adjacent vertices are at distance 1), and more importantly *they share no common neighbour*

in G: if some w were adjacent to both u and v in G, we'd have the path u-w-v of length 2, forcing $d_G(u,v) \leq 2$, contradicting $d_G(u,v) \geq 3$.

Now take *any* other vertex $x \in V(G) \setminus \{u,v\}$. Since u, v have no common G-neighbour, x cannot be adjacent to *both* u and v in G — so in $\overline{G}$, x is adjacent to *at least one* of u, v. Also, since u, v are non-adjacent in G, they *are* adjacent in $\overline{G}$.

We now show every pair of vertices is within distance 3 in $\overline{G}$:

- For u, v themselves: $d_{\overline{G}}(u,v) = 1$ (just shown).
- For u (or v) and any other vertex x: if x is $\overline{G}$-adjacent to u, done ($d = 1$); otherwise x is $\overline{G}$-adjacent to v (shown above), and v is $\overline{G}$-adjacent to u, giving the length-2 path x-v-u.
- For two other vertices x, y (neither equal to u, v): each of x, y is $\overline{G}$-adjacent to at least one of u, v. If they share one (say both adjacent to u in $\overline{G}$), then x-u-y has length 2. If not — say x is $\overline{G}$-adjacent to u only and y is $\overline{G}$-adjacent to v only — then x-u-v-y is a walk of length 3 in $\overline{G}$ (using the edge $uv \in \overline{G}$ from above), so $d_{\overline{G}}(x,y) \leq 3$.

Every pair of vertices is therefore within distance 3 in $\overline{G}$ (and in particular $\overline{G}$ is connected), so $\operatorname{diam}(\overline{G}) \leq 3$. ■

Note

This is a one-way statement, not an iff — the converse is false. Check $\operatorname{diam}(\overline{G}) \leq 3$ does *not* force $\operatorname{diam}(G) \geq 3$: take $G = K_n$ itself, $\operatorname{diam}(K_n) = 1$, while $\overline{K_n}$ has no edges at all so $\operatorname{diam}(\overline{K_n})$ is undefined (disconnected) — the implication is only ever claimed in the one direction stated.

2 What Is a Tree?

Definition (Tree)

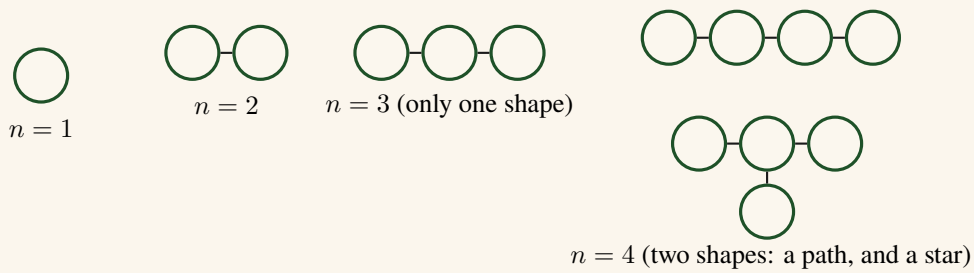
A **tree** is a connected graph that contains no circuit (no cycle). Equivalently, a tree is an **acyclic connected graph**.

Note

A tree is automatically a *simple* graph: a self-loop is a circuit of length 1, and a pair of parallel edges forms a circuit of length 2 — both are forbidden by "no circuit."

Example — Trees With 1, 2, 3, 4 Vertices

Every graph needs at least one vertex, so the smallest tree is a single vertex with no edges. Here are *all* trees (up to isomorphism) on 1, 2, 3, and 4 vertices:



Note that for $n = 3$ there's only *one* possible tree shape (a path), but for $n = 4$ there are two: the path P_4, and the star $K_{1,3}$. The number of distinct tree shapes grows quickly with n — for $n = 6$ there are already six, shown next.

Definition (Star, Path, Double Star)

Let T be a tree on n vertices.

- T is a **star**, $K_{1,n-1}$, if it has one vertex adjacent to all $n-1$ others (and those $n-1$ have degree 1).
- T is a **path**, P_n, if every vertex has degree ≤ 2 (so exactly two vertices — the ends — have degree 1, and the rest have degree 2).
- T is a **double star** if it has *exactly two* vertices that are not leaves, and (being a tree) those two must be adjacent.

Definition (Caterpillar)

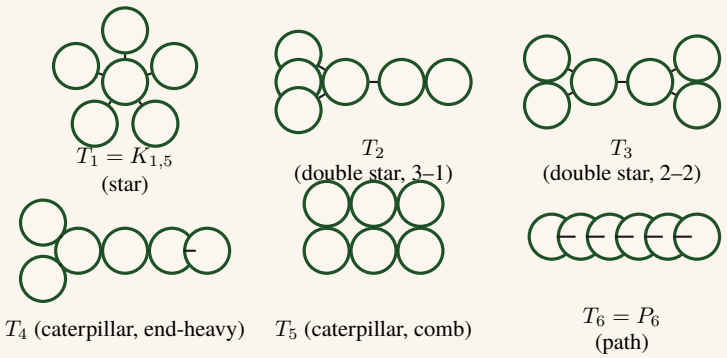
A **caterpillar** is a tree of order 3 or more in which deleting every leaf (every degree-1 vertex) leaves behind a path, called the **spine**.

Note

Every path, every star (of order ≥ 3), and every double star is automatically a caterpillar — in each case, once you prune the leaves, what's left is a path (possibly a path of length 0, i.e. a single vertex, or length 1, i.e. a single edge). Caterpillars are the next step up in complexity: a tree where all the "branching" happens along one central line.

Example — Six Trees of Order 6

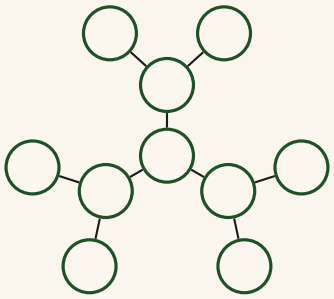
There are exactly six trees on 6 vertices, up to isomorphism (this is a fact you could in principle check by brute force; we won't prove it here). Here they are, arranged from "most spread out" (T_1) to "most stretched out" (T_6):



Note

$T_1 = K_{1,5}$ is a star and $T_6 = P_6$ is a path — the two extremes. T_2 and T_3 are double stars: exactly two non-leaf vertices (adjacent, as they must be), split 3–1 and 2–2. T_4 and T_5 are both caterpillars with a 3-vertex spine, but genuinely different trees: in T_4 the degree-3 vertex sits at one *end* of the spine (so one spine-vertex carries two leaves and the other carries one), while in T_5 the degree-3 vertex is the spine's *middle* vertex (one leaf on each of the three spine vertices, like a comb). Notice all six have the same number of edges, $5 = 6 - 1$, as Section 3's theorem demands — they differ only in *how* those 5 edges are arranged.

Example — A Tree That Is *Not* a Caterpillar



This "tripod" has center x joined to three branch vertices p_1, p_2, p_3, each of which carries two further leaves. Delete all six outer leaves (the degree-1 vertices) and what remains is $\{x, p_1, p_2, p_3\}$ with x joined to all three p_i — that's a *star*, not a path. So this tree is **not** a caterpillar: the branching happens at more than one place, not along a single spine.

Definition (Rooted Tree)

Sometimes it is convenient to single out one vertex of a tree T and call it the **root**. T, together with this choice, is then called a **rooted tree**. (A tree with no distinguished vertex is sometimes called a *free tree*, for contrast.)

3 The Big Equivalence Theorem: Six Ways to Say "Tree"

This is the heart of the chapter. The definition of a tree only says "connected + acyclic," but it turns out that — once you fix the number of vertices n — a whole list of other properties all say *exactly the same thing*. Knowing this list (and being able to prove it) is what lets you answer exam questions like "is G a tree?" using whichever piece of information you were actually handed (an edge count, a path-uniqueness claim, a bridge property, …) rather than only the original definition.

Warm-up — Check This on One Example First

Before the general proof, convince yourself of the claim on a tiny case. Take the path P_4 (4 vertices, 3 edges, no cycle):

- It's connected and acyclic. ✓ (condition (i))
- It's acyclic with $4 - 1 = 3$ edges. ✓ (condition (ii))
- It's connected with 3 edges. ✓ (condition (iii))
- Removing *any* one of its 3 edges disconnects it. ✓ (condition (iv))
- Every pair of its vertices is joined by exactly one path. ✓ (condition (v))
- It has no cycle, but adding any new edge (there are $\binom{4}{2} - 3 = 3$ possible new edges) creates exactly one cycle. ✓ (condition (vi))

All six hold together. Now take C_4 (a 4-cycle, same $n = 4$, but 4 edges, not 3): it fails (i) (not acyclic), fails (ii) and (iii) (wrong edge count), and every edge you remove leaves it *still connected* — so it fails (iv) too. The theorem below says this is not a coincidence: for a fixed n, all six conditions rise and fall together.

Definition (Bridge)

An edge e of a connected graph G is called a **bridge** if $G - e$ (the graph with e removed, vertices kept) is disconnected.

We will also need one small fact about paths before we can prove the theorem.

Lemma A — Two Distinct Paths Make a Cycle

If a graph contains two *distinct* paths between the same pair of vertices u and v, then it contains a cycle.

Proof. Let $P : u = x_0, x_1, \ldots, x_k = v$ and $Q : u = y_0, y_1, \ldots, y_l = v$ be two distinct u–v paths. Since $x_0 = y_0 = u$ but $P \neq Q$, there is a first index i where they disagree: let $i_0 \geq 0$ be the *largest* index with $x_{i_0} = y_{i_0}$ for all indices up to and including i_0 (so $i_0 \geq 0$ always works since $x_0 = y_0$), and call this shared vertex $w = x_{i_0} = y_{i_0}$, with $x_{i_0+1} \neq y_{i_0+1}$.

Walking onward from w along P and along Q, both paths eventually reach v again, so they must meet again at some common vertex after w (at worst, they meet at v itself). Let z be the *first* vertex after w at which they meet again — say $z = x_j$ on P and $z = y_{j'}$ on Q, with $j > i_0$, $j' > i_0$ chosen as small as possible.

Now consider the closed walk: go from w to z along P (through $x_{i_0+1}, \ldots, x_j$), then from z back to w along Q reversed (through $y_{j'-1}, \ldots, y_{i_0+1}$). Because P and Q are each paths (no repeated vertices *within* themselves), and because z was chosen as the *first* reconvergence after w, none of the intermediate vertices $x_{i_0+1}, \ldots, x_{j-1}$ coincides with any of $y_{i_0+1}, \ldots, y_{j'-1}$, and $w \neq z$ (since $x_{i_0+1} \neq y_{i_0+1}$ means P and Q genuinely separate right after w). So this closed walk revisits no vertex except at its own start/end point w — which is exactly the definition of a cycle. ■

Lemma B — A Connected Graph Needs at Least $n - 1$ Edges

Every connected graph on $n \geq 1$ vertices has at least $n - 1$ edges.

Proof (by induction on the number of edges m) We prove the more general statement: *a graph on n vertices with k connected components has at least $n-k$ edges.* (Taking $k = 1$, i.e. connected, gives the Lemma.)

Base case, $m = 0$. With no edges, every vertex is its own component, so $k = n$, and $n - k = 0 \leq m = 0$. ✓

Inductive step. Suppose the claim holds for every graph with fewer than m edges ($m \geq 1$). Let H have n vertices, k components, and m edges. Pick any edge e of H and delete it. $H - e$ has n vertices and $m - 1$ edges, so the induction hypothesis applies to it. Two cases:

- If e was *not* a bridge, $H - e$ still has k components, so by hypothesis $m - 1 \geq n - k$, giving $m \geq n - k + 1 > n - k$.
- If e *was* a bridge, $H - e$ has $k + 1$ components, so by hypothesis $m - 1 \geq n - (k + 1)$, giving $m \geq n - k$.

Either way $m \geq n - k$, completing the induction. ■

Main Theorem — Six Equivalent Conditions for a Tree

Let T be a graph with n vertices. The following six statements are all equivalent (each one implies all the others):

(i) T is a tree (connected and acyclic).

(ii) T is acyclic, and has $n - 1$ edges.

(iii) T is connected, and has $n - 1$ edges.

(iv) T is connected, and every edge of T is a bridge.

(v) Any two vertices of T are joined by *exactly one* path.

(vi) T is acyclic, but adding any single new edge to T creates exactly one cycle.

Proof. We prove (i)⇒(ii)⇒(iii)⇒(iv)⇒(v)⇒(vi)⇒(i). Going around this cycle of implications shows all six are equivalent: to get from any one of them to any other, just follow the arrows around.

(i)⇒(ii), by induction on n. Suppose T is connected and acyclic. *Base case* $n = 1$: no edges, and $n - 1 = 0$. ✓ *Inductive step*: suppose the claim holds for every tree with fewer than n vertices, and let T be a tree on $n \geq 2$ vertices. T has at least one edge (it's connected with ≥ 2 vertices). Pick any edge $e = uv$. We claim $T - e$ is disconnected: if it were still connected, it would contain some u–v path P not using e; but then P together with the edge e itself gives *two* distinct u–v connections in T (the path P, and the single edge e), so by Lemma A, T would contain a cycle — contradicting that T is acyclic. So $T - e$ is disconnected, and since deleting one edge can increase the number of components by at most one, $T - e$ has *exactly* two components, T_1 (containing u) and T_2 (containing v), on n_1 and n_2 vertices respectively, with $n_1 + n_2 = n$. Each T_i is connected (it's a component) and acyclic (a cycle in T_i would be a cycle in T too). So each T_i satisfies hypothesis (i), on fewer than n vertices, and by the induction hypothesis has $n_i - 1$ edges. Adding back the edge e:

$$|E(T)| = (n_1 - 1) + (n_2 - 1) + 1 = n_1 + n_2 - 1 = n - 1.$$

(ii)⇒(iii). Suppose T is acyclic with $n - 1$ edges, and suppose for contradiction T is disconnected, with $k \geq 2$ components $T_1, \ldots, T_k$ on $n_1, \ldots, n_k$ vertices ($\sum n_i = n$). Each T_i is connected (a component) and acyclic (inherited from T), so each T_i itself satisfies hypothesis (i) — and we *already proved* (i)⇒(ii) above, so each T_i has exactly $n_i - 1$ edges. Summing,

$$|E(T)| = \sum_{i=1}^{k} (n_i - 1) = n - k \leq n - 2,$$

since $k \geq 2$. This contradicts $|E(T)| = n - 1$. So T must be connected.

(iii)⇒(iv). Suppose T is connected with $n - 1$ edges. Let e be any edge; we must show $T - e$ is disconnected. Suppose instead $T - e$ were connected. $T - e$ has n vertices and $n - 2$ edges, so by Lemma B, a connected graph on n vertices needs $\geq n - 1$ edges — but $T - e$ only has $n - 2$. Contradiction. So $T - e$ is disconnected, i.e. e is a bridge. Since e was arbitrary, every edge of T is a bridge.

(iv)⇒(v). T is connected, so every pair of vertices has *at least* one path between them. Suppose some pair u, v had *two* distinct paths. By Lemma A, T would contain a cycle, say using edges $e_1, \ldots, e_r$ in a circuit. But every edge on a cycle fails to be a bridge: removing e_i from a cycle leaves the rest of the cycle as an alternate route between e_i's endpoints, so $T - e_i$ is still connected. This contradicts hypothesis (iv), which says *every* edge of T is a bridge. So no pair of vertices has two distinct paths: there is exactly one path between every pair.

(v)⇒(vi). First, T is acyclic: if T had a cycle, any two vertices on that cycle would be joined by two distinct paths (the two arcs of the cycle), contradicting uniqueness. Now let x, y be any two non-adjacent vertices of T (if T has no non-adjacent pair, $n \leq 2$ and there is nothing to add, so assume one exists). By hypothesis there is a unique x–y path P already in T. Adding the new edge xy creates the cycle "P followed by edge xy." Could it create a *second*, different cycle through xy? Any cycle through the new edge xy must consist of xy together with *some* x–y path already present in T — but P is the *only* such path, by hypothesis. So exactly one cycle is created.

(vi)⇒(i). T is already acyclic by hypothesis; we must show it is connected. Suppose not: let u, v lie in two different components of T. Add the edge uv. Since u and v were in different components, there was *no* u–v path in T before — so the new edge uv cannot be part of any cycle (a cycle through uv would need an old u–v path to close it up). So adding uv creates *zero* cycles, contradicting hypothesis (vi), which says adding any new edge creates exactly *one*. So T must already be connected. Combined with acyclic, T is a tree: condition (i). ■

Note

Condition (iv) is exactly the course slides' and Deo's notion of a **minimally connected graph** (a connected graph that disconnects as soon as you remove any one edge) — "every edge is a bridge" and

"minimally connected" are two names for the same property. So this theorem also proves, as one of its six equivalent faces, the corollary: *a connected simple graph is a tree if and only if it is minimally connected.*

4 Pendant Vertices

Definition (Pendant Vertex (Leaf))

A vertex of degree 1 in a graph is called a **pendant vertex**, or **leaf**.

Theorem — Every Tree Has at Least Two Leaves

Every tree T with $n \geq 2$ vertices has at least two pendant vertices.

Proof. By the Main Theorem (condition (iii)), T has $n-1$ edges. By the Handshaking Lemma, the sum of all degrees equals twice the number of edges:

$$\sum_{v \in V(T)} d(v) \;=\; 2|E(T)| \;=\; 2(n-1) \;=\; 2n-2.$$

Since T is connected with $n \geq 2$ vertices, *every* vertex has degree ≥ 1 (no isolated vertices). Suppose, for contradiction, that at most one vertex has degree 1 (i.e. T has zero or one leaf). Then at least $n-1$ of the n vertices have degree ≥ 2, so

$$\sum_{v \in V(T)} d(v) \;\geq\; 2(n-1) + 1 \cdot 1 \;=\; 2n-1 \quad \text{(if exactly one leaf)}, \quad \text{or} \quad \geq 2n \quad \text{(if zero leaves)},$$

either of which exceeds $2n-2$. This contradicts the exact total of $2n-2$ computed above. So T must have *at least two* vertices of degree 1. ■

Note

This proof is short precisely because it reuses a fact we already nailed down (the exact edge count $n-1$) instead of re-deriving everything from scratch by induction. This is the entire point of the Big Equivalence Theorem: once you know *one* characterisation holds, every other one is unlocked for free.

Lemma — Deleting a Leaf Keeps a Tree a Tree

If v is a leaf of a tree T with n vertices, then $T - v$ (delete v and its one incident edge) is a tree with $n-1$ vertices.

Proof. **Connected.** Let x, y be any two vertices of $T - v$ (so $x, y \neq v$). Since T is connected, there is a path P from x to y in T. Since v has degree 1 in T, v cannot be an *internal* vertex of any path between two other vertices (an internal vertex of a path has two distinct neighbours used by the path, but v has only one neighbour in all of T). So P does not pass through v, meaning P lies entirely within $T - v$. Hence $T - v$ is connected.

Acyclic. $T - v$ is a subgraph of T, and removing a vertex (with its edge) cannot create a cycle that wasn't already there. Since T has no cycle, neither does $T - v$.

So $T - v$ is connected and acyclic, i.e. a tree, on $n-1$ vertices. ■

Worked Example — Why This Matters: Longest Increasing Subsequences

Here is a genuinely useful application of trees (adapted from Deo), the kind of thing that shows up in programming interviews. Given the sequence $4, 1, 13, 7, 0, 2, 8, 11, 3$, we want the longest *monotonically*

increasing subsequence. Build a tree: one vertex per number, and draw an edge from number a to a later number b exactly when b continues an increasing run starting at a (in practice, you build this by scanning left to right and attaching each new number under the best vertex it can extend). Every root-to-vertex path in this tree spells out an increasing subsequence ending at that vertex, so the longest increasing subsequence corresponds to a *deepest leaf* in the tree — and there can be more than one, exactly when several leaves tie for maximum depth. For this sequence there turn out to be four longest increasing subsequences, each of length 4: $(4, 7, 8, 11)$, $(1, 7, 8, 11)$, $(1, 2, 8, 11)$, and $(0, 2, 8, 11)$. A tree built this way to represent data is sometimes called a **data tree**.

Exam Trap

"Every tree has at least two leaves" needs $n \geq 2$. The one-vertex tree ($n = 1$) has *zero* leaves (its only vertex has degree 0, not 1) — don't apply the theorem to it.

5 Forests

Definition (Forest)

A **forest** is an acyclic graph — equivalently, a graph in which every connected component is a tree. (A forest need not itself be connected; a single tree is the special case of a forest with exactly one component.)

Corollary — Edge Count of a Forest

Every forest F with n vertices and k connected components has exactly $n - k$ edges.

Proof. Let $G_1, \ldots, G_k$ be the connected components of F, with G_i having n_i vertices and m_i edges, so $n = \sum_{i=1}^{k} n_i$ and $|E(F)| = \sum_{i=1}^{k} m_i$. Each G_i, being a connected component of an acyclic graph, is itself connected and acyclic — a tree. By the Main Theorem, condition (iii), $m_i = n_i - 1$. Summing over all k components,

$$|E(F)| \;=\; \sum_{i=1}^{k} m_i \;=\; \sum_{i=1}^{k} (n_i - 1) \;=\; \Big(\sum_{i=1}^{k} n_i\Big) - k \;=\; n - k.$$

■

Example

Start with the 6-vertex caterpillar T_4 from Section 2 (5 edges, 1 component) and delete its middle spine edge. You get a forest with 6 vertices and 2 components: a 3-vertex star-shaped piece and a 3-vertex path-shaped piece. Edge count: $(3 - 1) + (3 - 1) = 4$. Check against the formula: $n - k = 6 - 2 = 4$. ✓

6 Distance and Centers — Specifically in Trees

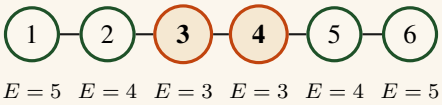
Section 1 defined distance, eccentricity, center, and radius for *any* connected graph. Trees make all of this much easier to compute, because of condition (v) of the Main Theorem: **there is exactly one path between any two vertices of a tree.** So in a tree, $d(u, v)$ is simply the length of *the* path between u and v — there's no competing shorter route to search for.

Worked Example — Eccentricities in a Small Tree

Take the path $T = P_6$ on vertices 1-2-3-4-5-6 (this is T_6 from Section 2). The eccentricity of each vertex is its distance to whichever end of the path is farther away:

vertex	1	2	3	4	5	6
$E(v)$	5	4	3	3	4	5

The minimum eccentricity is 3, achieved by *two* vertices (3 and 4) — so P_6 has **two centers**, and $\text{rad}(P_6) = 3$. The diameter (longest shortest-path) is $\text{diam}(P_6) = 5$ (between vertices 1 and 6).



Note

Notice $2 \times \text{rad}(P_6) = 6 \neq 5 = \text{diam}(P_6)$. Keep this example in mind — Theorem (Radius vs. Diameter) below uses exactly this fact.

Theorem (König) — Every Tree Has One or Two Centers

Every tree has either exactly one center, or exactly two centers; and if it has two, they are adjacent.

Proof (by induction on the number of vertices, via "peeling" leaves) **Key observation.** For any vertex v of a tree T, the farthest vertex from v is always *some leaf* of T (if the farthest vertex u had degree ≥ 2, the path from v to u could be extended one more step past u along u's other neighbour, which does not revisit v's side since T is a tree — producing a longer path and contradicting that u was farthest). So $E(v) = \max_i d(v, v_i)$ is always realised at a leaf.

Base cases. If T has 1 or 2 vertices, the single vertex (or either of the two, which are tied) is trivially the unique minimiser, or the unique pair, of eccentricity — T has one center ($n = 1$) or two adjacent centers ($n = 2$, the only edge's two endpoints, both with eccentricity 1).

Inductive step. Suppose the theorem holds for every tree with fewer vertices than T, and T has $n \geq 3$ vertices. By the leaf theorem above, T has at least two leaves. Let $T' = T$ with *all* its leaves deleted simultaneously. T' is still a tree (deleting leaves one at a time, by the Lemma of Section 4, always leaves a tree; doing this for every current leaf in turn never disconnects anything since each deletion only ever removes a degree-1 vertex).

Compare eccentricities in T and T': for any vertex w that survives into T', its farthest vertex in T was a leaf (by the key observation), and removing all leaves shortens the distance from w to every surviving vertex's old farthest-leaf-path by *exactly one step* (the last edge, into the now-deleted leaf, is gone, but the rest of the path survives since w itself is not a leaf and so is not deleted). So

$$E_{T'}(w) \;=\; E_T(w) - 1 \qquad \text{for every vertex } w \text{ of } T'.$$

Subtracting the same constant (1) from every surviving vertex's eccentricity does not change *which* vertices achieve the minimum. So the centers of T are exactly the centers of T' (both are literally the same set of vertices). Since T' has fewer vertices than T (as T had ≥ 2 leaves to remove), the induction hypothesis applies to T': it has one center, or two adjacent centers. Hence so does T. ■

Worked Example — Finding a Center by Peeling

Take the path P_7 on vertices $1, \ldots, 7$. Peel its two leaves (1 and 7): left with P_5 on $2, 3, 4, 5, 6$. Peel again (2 and 6): left with P_3 on $3, 4, 5$. Peel again (3 and 5): left with just vertex 4 — a single vertex, which must be the center. So $\text{center}(P_7) = \{4\}$, matching the direct computation $E(4) = 3$ (the smallest eccentricity in P_7).

Definition (Radius and Diameter of a Tree)

For a tree T: the **radius** $\text{rad}(T)$ is the eccentricity of its center(s); the **diameter** $\text{diam}(T)$ is the length of the longest path in T (equivalently, the largest eccentricity over all vertices).

Exam Trap — Radius Is *Not* Always Half the Diameter

It's tempting to guess $\text{diam}(T) = 2 \cdot \text{rad}(T)$ always. The P_6 example above is a ready-made counterexample: $\text{rad}(P_6) = 3$ but $\text{diam}(P_6) = 5 \neq 6$. What *is* always true (and worth knowing defensively) is the general inequality $\text{rad}(T) \leq \text{diam}(T) \leq 2 \cdot \text{rad}(T)$ — the diameter can fall short of exactly double the radius by at most 1, but it can fall short.

A Leadership-Selection Application

Suppose a 14-person society's communication links form a tree (every pair of people is reachable, but the network has no redundant link — lose any one link and someone is cut off). If "closeness to the farthest person" is what makes a good leader, the best choice is exactly a **center** of the tree: the center minimises the worst-case communication distance to anyone else in the group.

7 Rooted and Binary Trees

We met the *definition* of a rooted tree in Section 2. This section builds the vocabulary around it — the terminology you need for the "terminologies and properties" part of this chapter — and one especially important special case.

Definition (Level and Internal Vertex)

In a rooted tree T with root r: the **level** of a vertex v is its distance $d(r, v)$ from the root (so the root itself is at level 0). A vertex that is *not* a leaf is called an **internal vertex**.

Definition (Binary Tree)

A **binary tree** is a tree with three or more vertices in which exactly one vertex has degree 2, and every other vertex has degree 1 or 3.

Note

Since the degree-2 vertex is the unique vertex of its kind, it's natural to root the tree there — so every binary tree is automatically a rooted tree, rooted at its one degree-2 vertex. Each non-root vertex then has one edge "up" toward the root and (for internal vertices) two edges "down" — the familiar left-child/right-child picture.

Theorem — Two Basic Facts About Binary Trees

Let T be a binary tree on n vertices with p pendant (leaf) vertices and q internal (degree-3) vertices. Then:

(a) n is always odd.

(b) $q = p - 2$ — there are always exactly two *fewer* internal vertices than leaves.

Proof. **(a)** By definition, exactly one vertex has (even) degree 2, and the remaining $n - 1$ vertices all have odd degree (1 or 3). By the Handshaking Lemma's standard corollary, the number of odd-degree vertices in *any* graph is always even. So $n - 1$ is even, which forces n to be odd.

(b) Every vertex is one of: the single degree-2 vertex, a degree-1 (leaf) vertex, or a degree-3 (internal) vertex, so

$$n \;=\; 1 + p + q. \tag{$*$}$$

Counting edges by summing degrees (Handshaking Lemma) and using $|E(T)| = n - 1$ (Main Theorem, condition (iii)):

$$2(n-1) \;=\; 2\cdot 1 \;+\; 1\cdot p \;+\; 3\cdot q. \tag{$**$}$$

Substitute $(*)$ into the left side of $(**)$, i.e. replace n by $1 + p + q$:

$$2(p+q) \;=\; 2 + p + 3q \quad\Longrightarrow\quad 2p + 2q \;=\; 2 + p + 3q \quad\Longrightarrow\quad p \;=\; 2 + q \quad\Longrightarrow\quad q \;=\; p - 2.$$

■

Worked Example — A 13-Vertex Binary Tree

Solving $(*)$ and part (b) together for p in terms of n: substituting $q = p - 2$ into $n = 1 + p + q$ gives $n = 1 + p + (p-2) = 2p - 1$, so $p = \frac{n+1}{2}$ and $q = p - 2 = \frac{n-3}{2}$.
Take a 4-level binary tree with $n = 13$ vertices (root plus branching down to level 3). The formula predicts $p = \frac{13+1}{2} = 7$ leaves and $q = \frac{13-3}{2} = 5$ internal vertices, plus the 1 root — totalling $1 + 7 + 5 = 13$. ✓
Smallest case, $n = 3$: a single root (degree 2) with two children, both forced to be leaves (there are no other vertices left for them to connect to). So $p = 2$, $q = 0$. Check: $q = p - 2 = 0$. ✓

Why Binary Trees Matter — One-Sentence Pointers

Binary trees are the backbone of computer search procedures: each internal vertex is a yes/no test, each root-to-leaf path is a sequence of test outcomes, and reaching a particular leaf identifies the answer. The **height** of a binary tree (its largest level number) controls the worst-case number of tests; a **weighted path length** (each leaf weighted by how often it's the answer) controls the *average*-case number of tests — this is exactly the idea behind building efficient variable-length binary codes (e.g. Huffman-style codes) for data with uneven symbol frequencies. These ideas are foundational in algorithms courses; this course does not examine the optimisation algorithms themselves.

8 Two Advanced Results

We close with two results that go beyond what's needed for a first pass, but which your slides explicitly include — both have appeared, in sketch form, right at the end of the "Tree" section of your course material.

Theorem — Every Tree Embeds in a High-Minimum-Degree Graph

Let T be a tree of order n. If G is *any* graph with minimum degree $\delta(G) \geq n - 1$, then T is isomorphic to some subgraph of G.

Note

In plain words: if every vertex of G has enough neighbours ($\geq n-1$ of them), then *no matter what shape* the n-vertex tree T is, you can always find a copy of it sitting inside G. This is surprisingly strong — it doesn't matter whether T is a star, a path, or something lopsided; high minimum degree alone is enough.

Proof (by induction on n) **Base case, $n = 1$.** T is a single vertex. Any graph G (with $\delta(G) \geq 0$, trivially true) contains at least one vertex, so T embeds trivially.

Inductive hypothesis. Suppose that for every tree T' of order $k - 1$ (where $k \geq 2$) and every graph H with $\delta(H) \geq k - 2$, T' is isomorphic to some subgraph of H.

Inductive step. Let T have order $n = k$, and let G satisfy $\delta(G) \geq k - 1$. Let v be any leaf of T (exists by Section 4's theorem, since $k \geq 2$), and let u be v's unique neighbour in T. By the Lemma of Section 4, $T - v$ is a tree of order $k - 1$.

Since $\delta(G) \geq k - 1 > k - 2$, the inductive hypothesis applies with $H = G$: $T - v$ is isomorphic to some subgraph F of G. Let u' be the vertex of F corresponding to u under this isomorphism.

Now, u' has degree $\geq k - 1$ in G (since $\delta(G) \geq k - 1$), but F itself has only $k - 1$ vertices total, so u' has at most $k - 2$ *neighbours within* F (it can't be adjacent to itself). So u' must have at least one neighbour $w \in V(G) \setminus V(F)$ — a vertex of G lying *outside* the copy of $T - v$ we just found.

Attach w to F via the edge $u'w$. The resulting graph, F together with the new vertex w and edge $u'w$, is isomorphic to T: it has the same structure as $(T - v)$-plus-v-reattached-at-u, which is exactly T. This gives a subgraph of G isomorphic to T, completing the induction. ■

Worked Example — A Small Case

Let $T = K_{1,3}$ (a star on 4 vertices, so $n = 4$), and let $G = Q_3$, the 3-cube graph (8 vertices, every vertex of degree exactly 3, so $\delta(G) = 3 = n - 1$). The theorem guarantees T embeds in G — and indeed it does, trivially: pick *any* vertex v of Q_3 and its 3 neighbours; since every vertex of Q_3 has degree exactly 3, this is already a copy of $K_{1,3}$ sitting inside Q_3.

Definition (Wiener Index)

For a connected graph G, the **Wiener index** is

$$D(G) \;=\; \sum_{\{u,v\} \subseteq V(G)} d(u, v),$$

the sum of distances over all *unordered* pairs of distinct vertices. (Dividing by $\binom{n}{2}$, the number of pairs, gives the *average* pairwise distance — the Wiener index is just this average, left un-divided.)

Theorem — Stars Minimise, Paths Maximise

Among all trees on n vertices, the Wiener index $D(T)$ is minimised *uniquely* by the star $K_{1,n-1}$, and maximised *uniquely* by the path P_n.

Note

Quick sanity check with small cases (you can verify these by direct enumeration): for $n = 5$, every tree has $D(T)$ between $D(K_{1,4}) = 16$ and $D(P_5) = 20$; for $n = 6$, between $D(K_{1,5}) = 25$ and $D(P_6) = 35$. The star packs every pair as close together as possible (distance ≤ 2 always); the path spreads vertices as far apart as possible.

Proof (minimisation — the star) Let T be any tree on n vertices. T has $n - 1$ edges, so it has exactly $n - 1$ pairs of vertices at distance 1 (the edges themselves); *every other pair* is at distance ≥ 2. Hence

$$D(T) \;\geq\; (n-1) \cdot 1 \;+\; \left[\binom{n}{2} - (n-1)\right] \cdot 2 \;=\; (n-1) + 2\binom{n-1}{2},$$

since $\binom{n}{2} - (n-1) = \binom{n-1}{2}$ (a standard binomial identity: removing the $n - 1$ "distance-1" pairs from all $\binom{n}{2}$ pairs leaves exactly $\binom{n-1}{2}$). This lower bound is *achieved* exactly when every non-edge pair is at distance exactly 2 (no pair is ever farther than 2 apart) — and that happens precisely when T is a star: in $K_{1,n-1}$, the center is distance 1 from every leaf, and any two leaves are at distance exactly 2 (through the center), so indeed every pair is at distance 1 or 2. For the bound to be tight, we also need this to be the *only* way to achieve it: if T is not a star, some two leaves x, y fail to share a common neighbour realising distance 2 with every other pair simultaneously tight — more carefully, if T has diameter ≥ 3, some pair is at distance ≥ 3, which strictly

increases $D(T)$ past the bound. A tree has diameter ≤ 2 if and only if it has a vertex adjacent to all others, i.e. is a star. So the minimum is achieved uniquely by the star, with value

$$D(K_{1,n-1}) \;=\; (n-1) + 2\binom{n-1}{2} \;=\; (n-1) + (n-1)(n-2) \;=\; (n-1)^2.$$

■

Proof (maximisation — the path, by induction on n) **Base case,** $n = 1$**.** Trivial, $D(P_1) = 0$.

Setup. For a tree T and a leaf u of T, write $T - u$ for the (smaller) tree with u deleted. Splitting every pairwise distance in T into "pairs not involving u" and "pairs involving u":

$$D(T) \;=\; D(T-u) \;+\; \sum_{v \in V(T-u)} d(u,v).$$

Inductive hypothesis. Assume that among all trees on $n-1$ vertices, $D(\cdot)$ is uniquely maximised by P_{n-1}.

Inductive step. Let T be any tree on n vertices, and let u be any leaf of T. By the splitting identity above and the inductive hypothesis, $D(T-u) \leq D(P_{n-1})$, with equality iff $T-u$ is (isomorphic to) a path. It remains to bound $\sum_v d(u,v)$, the sum of distances from the leaf u to every other vertex.

Since T has $n-1$ other vertices, this sum is a sum of $n-1$ positive integers (all distances from u are ≥ 1, since $u \neq v$). Moreover, these $n-1$ distances cannot repeat "too much": in fact the sharpest possible spread — distances $1, 2, 3, \ldots, n-1$, all distinct, with no repeats and no gaps — is exactly what happens when u is the endpoint of a path and the tree has no other branching, and any repetition in the multiset of distances from u (which happens whenever T branches anywhere) can only *reduce* the sum compared to the gapless, repeat-free run $1, 2, \ldots, n-1$ for a fixed vertex count. (Intuitively: branching lets two different vertices share a shallow distance from u instead of one of them being pushed further out, which trades a large distance for a small one and lowers the total.) So

$$\sum_{v \in V(T-u)} d(u,v) \;\leq\; 1 + 2 + \cdots + (n-1) \;=\; \binom{n}{2},$$

with equality iff T is a path and u is one of its two endpoints. Combining both bounds,

$$D(T) \;=\; D(T-u) + \sum_{v} d(u,v) \;\leq\; D(P_{n-1}) + \binom{n}{2},$$

with equality iff T itself is P_n (both $T-u$ a path *and* u an endpoint of T forces $T = P_n$). This matches the direct recursive computation for paths, $D(P_n) = D(P_{n-1}) + \binom{n}{2}$, so the path achieves (and uniquely maximises) this bound. ■

Cheat Sheet — Day 11

Everything on One Page

- **Tree** = connected + acyclic. **Forest** = acyclic (possibly several trees).
- **Metric:** non-negative (zero iff equal), symmetric, triangle inequality. Graph distance $d(u,v)$ is always a metric on a connected graph.
- **Eccentricity** $E(v) = \max_x d(v,x)$. **Center=** min-eccentricity vertex(es). **Radius** = eccentricity of center. **Diameter** = max eccentricity.
- $\text{diam}(G) \geq 3 \implies \text{diam}(\overline{G}) \leq 3$ (one-way only).
- **Six equivalent ways to say "T (on n vertices) is a tree":** (i) connected + acyclic; (ii) acyclic + $(n-1)$

edges; (iii) connected + $(n-1)$ edges; (iv) connected + every edge a bridge ("minimally connected"); (v) exactly one path between every pair of vertices; (vi) acyclic, but any new edge creates exactly one cycle.

- **Lemma A:** two distinct paths between the same two vertices $\Rightarrow$ a cycle. **Lemma B:** connected on n vertices $\Rightarrow \geq n-1$ edges.
- Every tree with $n \geq 2$ has ≥ 2 leaves. Deleting a leaf keeps a tree a tree (one smaller).
- Forest, n vertices, k components $\Rightarrow$ exactly $n-k$ edges.
- **Every tree has 1 or 2 centers** (2 only if adjacent) — proved by peeling all leaves at once and inducting.
- $\text{rad}(T) \leq \text{diam}(T) \leq 2\,\text{rad}(T)$, but equality on the right can fail (e.g. P_6: rad 3, diam 5).
- **Binary tree:** exactly one degree-2 vertex, rest degree 1 or 3. n always odd; internal vertices $q = p-2$ where p = number of leaves.
- **Embedding:** tree of order n embeds in any G with $\delta(G) \geq n-1$.
- **Wiener index** $D(G) = \sum_{\{u,v\}} d(u,v)$. Among n-vertex trees: star *uniquely* minimises $(= (n-1)^2)$, path *uniquely* maximises.

Practice Questions

Q1

[Original] A graph G has 9 vertices. For each of the following, say whether G *must* be a tree, giving a one-line reason: (a) G is connected with 8 edges and no cycle. (b) G has 8 edges and no cycle, but you are not told it is connected. (c) G is connected and every edge is a bridge. (d) G has 8 edges and is connected.

Q2

[Course slides – Homework, Tree section] Let T be a tree with $n \geq 4$ vertices, and let v be a vertex of *maximum* degree in T. Prove that if $d(v) = 2$, then T is a path.

Q3

[Deo – 3-6, adapted] Draw a tree in which the radius is *not* half the diameter, and compute both values. (You may reuse a tree from this chapter, but explain why it works, in your own words.)

Q4

[Original] A forest has 23 vertices and 4 connected components. How many edges does it have? If you're additionally told one component is a single isolated vertex and another is a path on 5 vertices, how many vertices are in the remaining two components combined?

Q5

[Original] Let T be the caterpillar with spine a-b-c-d (4 spine vertices) and exactly one extra leaf attached to each of b and c (so T has 6 vertices total). (a) Find all the leaves of T. (b) Compute the eccentricity of every vertex, and identify the center(s). (c) Is T a caterpillar? Justify from the definition.

Q6

[Original] True or false, with justification: "Every caterpillar is a double star." Then: "Every double star is a caterpillar."

Q7

[Deo – 3-2, adapted] A binary tree has 11 leaves. How many vertices does it have in total, and how many are internal (degree-3)?

Q8

[Wilson – Ch.3, in the spirit of Thm 3.1] Prove or disprove: a connected graph G on n vertices with exactly n edges (one *more* than a tree) must contain exactly one cycle.

Q9

[Original] Compute the Wiener index $D(T)$ by hand for the double star T_3 (Section 2: two adjacent centers, each with 2 leaves, 6 vertices total). Compare it against the star bound $(n-1)^2$ and the path bound for $n = 6$, and confirm it lies strictly between them.

Q10

[Original] Let G be a graph on 7 vertices with $\delta(G) \geq 6$. What is G, necessarily? Using this, explain why the Tree Embedding Theorem is "obviously" true in this particular extreme case — and why the theorem only becomes interesting when G has *more* than n vertices.

The Next Three — Closer to Exam Style

Everything above checks whether you can *execute* a proof technique or formula on a given instance. The next three are closer to how your actual Mid-Sem was written: a true/false or "construct an example" question that forces you to *combine* tools from this chapter (and earlier ones) in a way no single theorem here hands you directly.

Q11

[Wilson – Ch.3, Exercise 3.4] Prove that every tree is a bipartite graph. Then: which trees are *complete* bipartite graphs ($K_{m,k}$ for some m, k)?

Q12

[Wilson – Ch.3, Exercise 3.10] Construct two explicit trees, each on 7 vertices: one with exactly one center, and one with exactly two (necessarily adjacent) centers. Justify each by computing eccentricities.

Q13

[Original] Prove: if v is a vertex of degree Δ in a tree T, then T has *at least* Δ leaves. Then find a tree where this inequality is *strict* (more leaves than the maximum degree) — you've already built one earlier in this chapter.

Solutions

Solution

Q1. All four: **Yes, G must be a tree** — but (b) is the interesting one.
(a) Connected, acyclic, $8 = n - 1$ edges: this literally states both condition (ii) and condition (iii) of the Main Theorem at once, so certainly a tree.
(b) **Perhaps surprisingly, still yes** — even though you are *not* told G is connected. "Acyclic with $n - 1$ edges" is exactly condition (ii) of the Main Theorem, and the proof of (ii)⇒(iii) shows this already *forces* connectedness: a disconnected acyclic graph on 9 vertices with $k \geq 2$ components has only $9 - k \leq 7$ edges (Section 3's component-counting argument), never 8. So an acyclic 9-vertex graph with 8 edges cannot be disconnected — the edge count alone rules it out. This is the whole point of condition (ii): it looks like it's missing "connected," but it isn't.
(c) Connected with every edge a bridge is condition (iv) directly.
(d) Connected with $8 = n - 1$ edges is condition (iii) directly.

Solution

Q2. Since v has maximum degree in T and $d(v) = 2$, *every* vertex of T has degree ≤ 2.
Since $n \geq 4 \geq 2$, T has at least two leaves (Section 4). In particular T is not 2-regular (some vertex has degree exactly 1), so not every vertex has degree exactly 2.
Pick any leaf x_0. It has one neighbour x_1. Since $n \geq 4$, $T \neq$ a single edge, so x_1 is not also a leaf (if it were, connectivity would force $n = 2$); so x_1 has degree exactly 2 (it can't exceed the max degree 2, and isn't 1), meaning it has exactly one neighbour besides x_0 — call it x_2, necessarily different from x_0 (else we'd have a cycle $x_0x_1x_0$, impossible with simple edges, or more generally revisiting any earlier vertex would create a cycle, contradicting that T is acyclic). Continue this way: at each new vertex x_i ($i \geq 1$) of degree 2, there is exactly one "next" neighbour not yet visited (using the same no-revisit argument), until we reach a vertex x_k of degree 1 — a leaf, where the chain stops (this must happen in at most n steps, since there are only n vertices and we never repeat one).
This produces a path $P : x_0\text{-}x_1\text{-}\cdots\text{-}x_k$ inside T. We claim P already uses *every* vertex of T. Suppose not: since T is connected, some vertex off P must be adjacent to some vertex *on* P (take any vertex $y \notin P$; a T-path from y to x_0 must cross from off-P to on-P somewhere). But every internal vertex x_i ($0 < i < k$) already has both its degree-2 slots used up *within* P (one edge to x_{i-1}, one to x_{i+1}), and both endpoints x_0, x_k are leaves with their one slot already used (the edge into P) — so *no* vertex of P has room for an extra neighbour outside P. This contradicts the connectivity argument. So P must include all n vertices, i.e. $T = P$, a path. ■
(Shortcut, once you trust the definitions: maximum degree 2 means *every* vertex has degree ≤ 2, which is exactly this chapter's definition of a path — so the statement is almost immediate from the definitions alone. The longer argument above is what that one-line shortcut is really saying underneath, spelled out in full.)

Solution

Q3. Use P_6 (path on 6 vertices, Section 2's T_6, and Section 6's worked example). Its eccentricities are $5, 4, 3, 3, 4, 5$, so $\text{rad}(P_6) = 3$ (two centers, vertices 3 and 4) and $\text{diam}(P_6) = 5$. Half the diameter would be 2.5, which isn't even an integer, let alone equal to 3 — so radius $\neq$ half the diameter here. This happens because P_6 has an *even* number of vertices: the "middle" of the path is an edge (between the two centers), not a single vertex, which is what breaks the clean $\text{diam} = 2 \cdot \text{rad}$ relationship that odd-length paths (like P_5, P_7) happen to satisfy.

Solution

Q4. By the Forest Corollary, edges $= n-k = 23-4 = 19$. The isolated vertex (1 vertex, 0 edges, its own component) and the 5-vertex path (5 vertices) account for $1+5=6$ vertices in 2 of the 4 components, leaving $23-6=17$ vertices split between the remaining 2 components.

Solution

Q5. Vertices: spine a, b, c, d plus leaves b', c' (attached to b, c respectively).
(a) Leaves (degree 1): a, d, b', c'. (b and c have degree 3 each: two spine neighbours plus one leaf.)
(b) Distances from a: a-b:1, a-c:2, a-d:3, a-b':2, a-c':3. So $E(a) = 3$. By the symmetry of this tree (swap $a \leftrightarrow d$, $b \leftrightarrow c$, $b' \leftrightarrow c'$), $E(d) = 3$ too. From b: b-a:1,b-c:1,b-d:2,b-b':1,b-c':2, so $E(b) = 2$; by symmetry $E(c) = 2$. From b': b'-a:2,b'-b:1,b'-c:2,b'-d:3,b'-c':3, so $E(b') = 3$; by symmetry $E(c') = 3$. Minimum eccentricity is 2, achieved by b and c only — so **center** $= \{b, c\}$ (adjacent, as the theorem guarantees).
(c) **Yes.** Removing the 4 leaves $\{a, d, b', c'\}$ leaves exactly $\{b, c\}$ with the edge bc — a path (of length 1) — so by definition T is a caterpillar with spine b-c.

Solution

Q6. "Every caterpillar is a double star": False. A caterpillar can have a spine of *any* length, e.g. the spine-of-3 caterpillars T_4, T_5 from Section 2 have *three* non-leaf vertices, not two, so they are caterpillars but not double stars.
"Every double star is a caterpillar": True. A double star has exactly two non-leaf (adjacent) vertices; removing all its leaves leaves exactly those two vertices joined by an edge — a path of length 1 — satisfying the definition of caterpillar directly (this was already noted as a Remark in Section 2).

Solution

Q7. Using $q = p - 2$ from Section 7 with $p = 11$: $q = 11 - 2 = 9$ internal vertices. Total vertices $n = 1 + p + q = 1 + 11 + 9 = 21$ (check: odd, as required).

Solution

Q8. True. Such a G has one edge more than a spanning tree would. Take any spanning tree T of G (exists since G is connected) using $n-1$ of G's edges; exactly one edge e of G is left over. Adding e back to T is exactly the situation of Main Theorem condition (vi): T was acyclic, and adding the single new edge e creates *exactly one* cycle. So $G = T + e$ contains exactly one cycle — no more, since every other edge of G is already in the acyclic T, and no combination of T's edges alone can close a cycle (that's precisely acyclicity).

Solution

Q9. Label T_3's two centers a, b (adjacent) with leaves a_1, a_2 (on a) and b_1, b_2 (on b). Distances: $d(a,b) = 1$ (1 pair); $d(a,a_1) = d(a,a_2) = d(b,b_1) = d(b,b_2) = 1$ (4 pairs); $d(a,b_1) = d(a,b_2) = d(b,a_1) = d(b,a_2) = 2$ (4 pairs); $d(a_1,a_2) = 2$ (1 pair); $d(b_1,b_2) = 2$ (1 pair); $d(a_1,b_1) = d(a_1,b_2) = d(a_2,b_1) = d(a_2,b_2) = 3$ (4 pairs). Total: $(1+4)\cdot 1 + (4+1+1)\cdot 2 + 4\cdot 3 = 5 + 12 + 12 = 29$. Check against bounds for $n = 6$: star gives 25, path gives 35, and indeed $25 < 29 < 35$. ✓

Solution

Q10. With $n = 7$ vertices and $\delta(G) \geq 6 = n - 1$, every vertex is adjacent to *all* other 6 vertices — so G must be K_7, the complete graph. The Embedding Theorem is "obvious" here because K_7 already contains *every* graph (and in particular every tree) on 7 vertices as a subgraph, trivially — there's no structure to find, since all possible edges are already present. The theorem becomes genuinely interesting

once G has *more* vertices than T (as in the Q_3 example in Section 8): then G is *not* complete, and the claim that a copy of T can still always be found, purely from a minimum-degree bound, is a real structural guarantee rather than a triviality.

Solution

Q11. Every tree is bipartite. Let T be a tree; root it at any vertex r. For a vertex v, define its **level** $\ell(v) = d(r, v)$ (well-defined since T has a unique r–v path). *Claim: every edge of T joins a vertex at some level i to a vertex at level $i + 1$* — never two vertices at the *same* level. Indeed, let uv be an edge with, say, $\ell(u) \leq \ell(v)$. Extending the unique r–u path by the edge uv gives *an* r–v walk of length $\ell(u) + 1$; since T has only *one* path from r to v (Main Theorem, condition (v)), this walk, if it doesn't repeat a vertex, must in fact be the unique r–v path itself (two different simple r-v routes would violate uniqueness) — forcing $\ell(v) = \ell(u) + 1$ exactly. So consecutive levels always differ by exactly 1 across any edge.

Now 2-colour T: colour every even-level vertex red, every odd-level vertex blue. By the claim, every edge joins a red vertex to a blue one — never same-to-same. That's exactly a proper 2-colouring, so T is bipartite, with parts $\{\text{even levels}\}$ and $\{\text{odd levels}\}$.

Which trees are complete bipartite? Exactly the **stars** $K_{1,n-1}$. If both parts of $K_{m,k}$ have ≥ 2 vertices, pick a_1, a_2 from one side and b_1, b_2 from the other: a_1, b_1, a_2, b_2 already forms a 4-cycle a_1-b_1-a_2-b_2-a_1, so $K_{m,k}$ is *not* acyclic unless one part has only 1 vertex — i.e. $K_{1,n-1}$, the star.

Solution

Q12. One center: P_7 (path on 7 vertices), already computed in Section 6's worked example by peeling: center $= \{4\}$ (the middle vertex), with eccentricities $6, 5, 4, 3, 4, 5, 6$ reading along the path — a single, strict minimum at the middle.

Two centers: the double star with split 3–2: two adjacent centers a, b, with a carrying 3 leaves and b carrying 2 leaves ($2+3+2 = 7$ vertices total). Eccentricities: from a, the farthest vertices are b's leaves, at distance 2 (a-b-leaf), so $E(a) = 2$; by the same reasoning $E(b) = 2$. Any leaf x of a, though, is at distance 3 from any leaf of b (leaf-a-b-leaf), so $E(x) = 3$, and symmetrically every leaf (on either side) has eccentricity 3. Minimum eccentricity is 2, achieved only by a and b — two centers, adjacent, as the theorem requires.

Solution

Q13. Leaves $\geq \Delta$. Let v have degree Δ in T, with neighbours $w_1, \ldots, w_\Delta$. Removing v splits T into exactly Δ components $T_1, \ldots, T_\Delta$, one containing each w_i — because for $i \neq j$, the unique w_i–w_j path in T must pass through v (otherwise that path together with edges vw_i, vw_j would close a cycle through v, impossible in a tree), so w_i, w_j land in different components once v is removed.

Each T_i contributes *at least one* leaf of T: if $T_i = \{w_i\}$ alone, then w_i's only edge in all of T was to v, so w_i itself is a leaf of T. If T_i has more vertices, T_i is itself a tree on ≥ 2 vertices, so it has ≥ 2 leaves *within* T_i (Section 4); at most one of those can be w_i (whose degree in T is its T_i-degree *plus one*, for the edge up to v, so w_i is never a leaf of T once it has any T_i-neighbour). So T_i has some leaf-of-T_i other than w_i — and since that vertex has no edge to v, its degree is the same in T_i and in T, making it a genuine leaf of T.

Different components are disjoint, so these Δ contributed leaves are all distinct: T has at least Δ leaves.

Strict inequality: the double star T_3 (Section 2: split 2–2) has maximum degree $\Delta = 3$ (each center has 2 leaves $+ 1$ edge to the other center), but 4 leaves total — $4 > 3 = \Delta$. (The bound *can* be tight too: the star $K_{1,n-1}$ has $\Delta = n - 1$ and exactly $n - 1$ leaves.)